```
!pip install -q ultralytics
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=Tr
```

```
!nvidia-smi
```

```
!unzip -q data.zip -d /content/raw
!ls /content/raw
```

```
Ds1  Ds2  Ds3
```

```
import yaml
from pathlib import Path
RAW = Path("/content/raw")
if not (RAW/"Ds1").exists() and (RAW/"raw"/"Ds1").exists():
    RAW = RAW/"raw"
for ds in ["Ds1","Ds2","Ds3"]:
    names = yaml.safe_load(open(RAW/ds/"data.yaml"))["names"]
    print(ds, names)
```

```
Ds1 ['batteries', 'can', 'electronics', 'others', 'paper', 'pet_bottle', 'plastic_other', 'wet_waste']
Ds2 ['Cardboard', 'Glass', 'Metal', 'Paper', 'Plastic']
Ds3 ['Aerosols', 'Aluminum can', 'Aluminum caps', 'Cardboard', 'Cellulose', 'Ceramic', 'Combined plastic', 'Container for hc
```

```
import yaml, shutil, re
from pathlib import Path

RAW = Path("/content/raw")
if not (RAW/"Ds1").exists() and (RAW/"raw"/"Ds1").exists():
    RAW = RAW/"raw"
OUT = Path("/content/combined")
if OUT.exists(): shutil.rmtree(OUT)

TARGET = ["paper","cardboard","plastic","plastic_bag","glass","metal","battery","aerosol"]
tid = {n: i for i, n in enumerate(TARGET)}

# first matching rule wins, so the order matters
RULES = [
    (r"aerosol", "aerosol"),
    (r"batter", "battery"),
    (r"paper bag", "paper"),
    (r"plastic bag|\bbag\b", "plastic_bag"),
    (r"glass", "glass"),
    (r"cardboard|carton", "cardboard"),
    (r"paper", "paper"),
    (r"metal|aluminu?m|\bcans?\b|\btin\b|steel", "metal"),
    (r"plastic|bottle|\bpet\b", "plastic"),
]
def target_of(name):
    n = re.sub(r"[_\-]+", " ", name.lower()).strip()
    for pat, t in RULES:
        if re.search(pat, n): return t
    return None

kept = {n: 0 for n in TARGET}
for ds in ["Ds1","Ds2","Ds3"]:
    names = yaml.safe_load(open(RAW/ds/"data.yaml"))["names"]
    if isinstance(names, dict): names = [names[i] for i in sorted(names)]
    old2new = {}
    for i, n in enumerate(names):
        t = target_of(n)
        print(f"{ds}: {n!r} -> {t}")
        if t: old2new[i] = tid[t]
    for split in ["train","valid","test"]:
        imgs = RAW/ds/split/"images"
        if not imgs.exists(): continue
        (OUT/split/"images").mkdir(parents=True, exist_ok=True)
        (OUT/split/"labels").mkdir(parents=True, exist_ok=True)
        for img in imgs.iterdir():
            lbl = RAW/ds/split/"labels"/(img.stem + ".txt")
            lines = []
            if lbl.exists():
                for ln in lbl.read_text().splitlines():
                    p = ln.split()
                    if len(p) == 5 and int(p[0]) in old2new:   # boxes only
```

```
                        lines.append(" ".join([str(old2new[int(p[0])])] + p[1:]))
                        kept[TARGET[old2new[int(p[0])]]] += 1
            if not lines: continue
            shutil.copy(img, OUT/split/"images"/f"{ds}_{img.name}")
            (OUT/split/"labels"/f"{ds}_{img.stem}.txt").write_text("\n".join(lines))

yaml.dump({"path": str(OUT), "train": "train/images", "val": "valid/images",
           "test": "test/images", "names": TARGET}, open(OUT/"data.yaml","w"))
print("\nBoxes kept per class:", kept)
```

```
Ds1: 'batteries' -> battery
Ds1: 'can' -> metal
Ds1: 'electronics' -> None
Ds1: 'others' -> None
Ds1: 'paper' -> paper
Ds1: 'pet_bottle' -> plastic
Ds1: 'plastic_other' -> plastic
Ds1: 'wet_waste' -> None
Ds2: 'Cardboard' -> cardboard
Ds2: 'Glass' -> glass
Ds2: 'Metal' -> metal
Ds2: 'Paper' -> paper
Ds2: 'Plastic' -> plastic
Ds3: 'Aerosols' -> aerosol
Ds3: 'Aluminum can' -> metal
Ds3: 'Aluminum caps' -> metal
Ds3: 'Cardboard' -> cardboard
Ds3: 'Cellulose' -> None
Ds3: 'Ceramic' -> None
Ds3: 'Combined plastic' -> plastic
Ds3: 'Container for household chemicals' -> None
Ds3: 'Disposable tableware' -> None
Ds3: 'Electronics' -> None
Ds3: 'Foil' -> None
Ds3: 'Furniture' -> None
Ds3: 'Glass bottle' -> glass
Ds3: 'Iron utensils' -> None
Ds3: 'Liquid' -> None
Ds3: 'Metal shavings' -> metal
Ds3: 'Milk bottle' -> plastic
Ds3: 'Organic' -> None
Ds3: 'Paper' -> paper
Ds3: 'Paper bag' -> paper
Ds3: 'Paper cups' -> paper
Ds3: 'Paper shavings' -> paper
Ds3: 'Papier mache' -> None
Ds3: 'Plastic bag' -> plastic_bag
Ds3: 'Plastic bottle' -> plastic
Ds3: 'Plastic can' -> metal
Ds3: 'Plastic canister' -> plastic
Ds3: 'Plastic caps' -> plastic
Ds3: 'Plastic cup' -> plastic
Ds3: 'Plastic shaker' -> plastic
Ds3: 'Plastic shavings' -> plastic
Ds3: 'Plastic toys' -> plastic
Ds3: 'Postal packaging' -> None
Ds3: 'Printing industry' -> None
Ds3: 'Scrap metal' -> metal
Ds3: 'Stretch film' -> None
Ds3: 'Tetra pack' -> None
Ds3: 'Textile' -> None
Ds3: 'Tin' -> metal
Ds3: 'Unknown plastic' -> plastic
Ds3: 'Wood' -> None
Ds3: 'Zip plastic bag' -> plastic_bag

Boxes kept per class: {'paper': 925, 'cardboard': 1072, 'plastic': 2386, 'plastic_bag': 940, 'glass': 1053, 'metal': 2708,
```

```
!yolo detect train data=/content/combined/data.yaml model=yolo11n.pt epochs=40 imgsz=640 batch=32 patience=10 project=/conte
```

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.3it/s
```

```
                 all       1351       1787      0.583      0.647      0.624      0.418

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      33/40      6.45G      0.989      0.812      1.461         21        640: 100% ━━━━━━━━━━━━ 144/144 2.9it/s 49.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.2it/s
                   all       1351       1787      0.725      0.601       0.66      0.451

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      34/40      6.45G     0.9689     0.7926      1.445         16        640: 100% ━━━━━━━━━━━━ 144/144 3.0it/s 48.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.2it/s
                   all       1351       1787      0.771      0.567      0.625      0.413

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      35/40      6.45G     0.9572     0.7836       1.43         16        640: 100% ━━━━━━━━━━━━ 144/144 3.0it/s 48.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.2it/s
                   all       1351       1787      0.674      0.598       0.64      0.427

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      36/40      6.45G     0.9557     0.7525      1.419         20        640: 100% ━━━━━━━━━━━━ 144/144 2.9it/s 49.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.1it/s
                   all       1351       1787      0.668      0.591      0.639      0.428

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      37/40      6.45G     0.9395     0.7257       1.41         29        640: 100% ━━━━━━━━━━━━ 144/144 2.9it/s 49.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.1it/s
                   all       1351       1787      0.721      0.554      0.642      0.434

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      38/40      6.45G     0.9272     0.7214      1.401         18        640: 100% ━━━━━━━━━━━━ 144/144 2.9it/s 49.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.2it/s
                   all       1351       1787      0.657      0.619      0.653      0.442

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      39/40      6.45G     0.9204     0.6958      1.392         17        640: 100% ━━━━━━━━━━━━ 144/144 3.0it/s 48.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 22/22 2.2it/s
                   all       1351       1787       0.67      0.612      0.647      0.436

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      40/40      6.45G     0.9082     0.6781      1.384         16        640: 100% ━━━━━━━━━━━━ 144/144 3.0it/s 48.4s
```

```
!yolo detect predict model=/content/drive/MyDrive/smart-mrf-runs/v1/weights/best.pt source=/content/combined/valid/images sa
```

```
image 1213/1351 /content/combined/valid/images/Ds3_d8aec2be-AluCan258_jpg.rf.5124495bd122243681be48169b135049.jpg: 640x480
image 1214/1351 /content/combined/valid/images/Ds3_d94ddb08-R_924_jpg.rf.d82322e2127e268d4e8c74662dd420df.jpg: 640x640 1 p
image 1215/1351 /content/combined/valid/images/Ds3_d9581e93-AluCan833_jpg.rf.ff88d87e3e5bf55b2a70f89136034ddb.jpg: 640x480
image 1216/1351 /content/combined/valid/images/Ds3_d9618871-AluCan625_jpg.rf.8ab2f2da400be2a164ca642a8221a189.jpg: 640x480
image 1217/1351 /content/combined/valid/images/Ds3_d99d7b02-R_3952_jpg.rf.6e117974a0c0ced1ad6d2ca8ffb93305.jpg: 640x192 1
image 1218/1351 /content/combined/valid/images/Ds3_d9ba50d8-R_3376_jpg.rf.5ee07bb3506dce762dc9e6e2bf8ea5f1.jpg: 480x640 3
image 1219/1351 /content/combined/valid/images/Ds3_d9da18d8-AluCan3_jpg.rf.22dc8a391df1e13480d5f5994c8b43eb.jpg: 480x640 1
image 1220/1351 /content/combined/valid/images/Ds3_da736e86-R_413_jpg.rf.35cc1038fd44fbf0e8ce481b67f9f448.jpg: 384x640 2 p
image 1221/1351 /content/combined/valid/images/Ds3_dab14c2f-R_3909_jpg.rf.2212363e4e774720d76b95a30da57a7d.jpg: 640x480 1
image 1222/1351 /content/combined/valid/images/Ds3_dac15356-AluCan1058_jpg.rf.1b2a60630f8e312e45158130e263003f.jpg: 640x48
image 1223/1351 /content/combined/valid/images/Ds3_dba70117-AluCan250_jpg.rf.f26e05f6a37ae0e2ea180c11abe8fe0f.jpg: 640x480
```

```
import glob
from IPython.display import Image, display
for p in glob.glob('/content/runs/detect/predict/*.jpg')[:10]:
    display(Image(filename=p, height=400))
```